# Algebraic Numbers and Functions Course Notes

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## 1 Galois review

**Definition 1.1.** Seperable means equivalently

1. the number of distinct $k$-embeddings $L \hookrightarrow K$ is $[L : K]$
2. For $\alpha \in L$, the minimal polynomial of $\alpha$ over $K$ is "seperable": having no repeated roots in $\overline{K}$, i.e. $\gcd(f(x), f'(x)) = 1$.

**Definition 1.2.** Normal means equivalently

1. every $k$-embedding $L \hookrightarrow \overline{K}$ has the same image
2. $L$ is a splitting field (of a family of polynomials).
3. Given an irreducible polynomial $f(x) \in K[x]$, either has no roots in $L$ or all of its roots are in $L$.

**Definition 1.3.** $K \subseteq L$ is Galois if it is separable and normal.

**Definition 1.4.**

$$\mathrm{Gal}(L/K) := \{\sigma : L \to L \text{ automorphisms s.t. } \sigma|_k = id_k\}$$

(fixing base $k$-embedding $L \hookrightarrow \overline{K}$), Gal is identified with k-embeddings $L \hookrightarrow \overline{K}$.

If $L$ is a finite Galois group over $K$, the Galois correspondence: subgroups

$$\{H \leq G\}$$

intermediate fields:

$$\{K \leq F \leq L\}$$

$H \mapsto L^H := \{\alpha \in L | \sigma(\alpha) = \alpha \text{ for } \sigma \in H\}$ and

$$F \mapsto \mathrm{Gal}(L/F)$$

$H \trianglelefteq G$ iff $F$ is Galois / $K$.
$|\mathrm{Gal}(L/K)| = [L : K]$.

## 2 PRime ideals

$P$ is a prime ideal. We have $P \subset R \subset K$. If we Galois extension $K$ to $L$, we have $Q \subset S \subset L$.

**Definition 2.1.** $P \subset Q$ means "$Q$ is a prime over $P$".

**Theorem 2.2.** *Theorem 19 of marcus.*

1. $P \subset Q \implies Q \cap R = P$.

**Theorem 2.3.** *Theorem 20 of marcus.*
*Every $P$ has exactly $\geq 1$ $Q$ and every $Q$ has exactly 1 $P$.*